

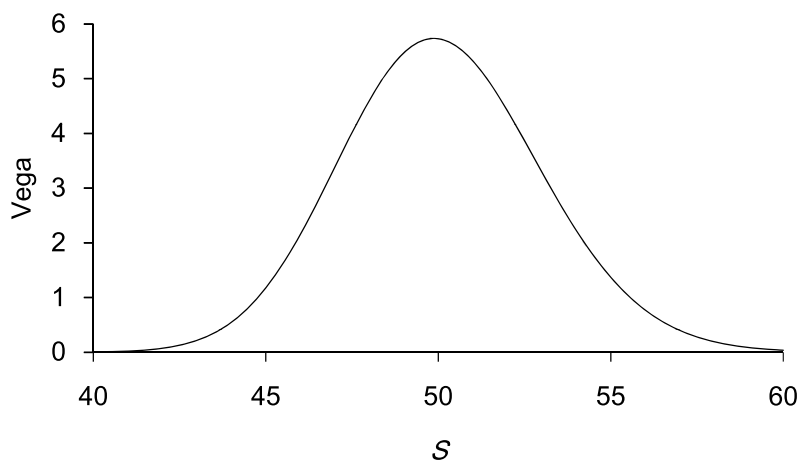


Figure A.14: Vega of a Call Option as a Function of Underlying Asset When $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

Figure A.14 shows the vega of a call option according to the Black-Scholes formula. If we multiply the vega by 0.01, this will tell us how much the value of the option will increase given a 1% increase in implied volatility. The vega of an option tends to be largest when the underlying is near the strike and falls off as we move away from the strike.

Whereas gamma is usually a measure that is associated with short-dated options, vega is a term that is usually associated with long-dated options. In the rates markets, for example, the terms “short-dated options” and “gamma” are often used interchangeably, and the terms “long-dated options” and “vega” are also synonymous, with the demarcation between the two usually at an expiry of one year. The reason for this is that whereas short-dated options can have large gamma, vega is typically quite small; an increase in implied volatility is not going to change the value of a short-dated option all that much. That is, option traders tend to trade short-dated options for their gamma profile, not their vega profile, and trade long-dated options for their vega profile, not their gamma profile. Option traders often buy short-dated options because of their significant gamma and the associated benefits from employing certain delta hedging strategies of these positions, which is why these options have come to be known as simply “gamma.” An option trader who wants to take a view on implied volatility would be much more likely to take a position in a long-dated option that can have significant exposure to changes in implied volatility. It’s for this reason that trading long-dated options has become known as trading “vega.”

A.6.4 Theta

The change in the value of an option given an increase in time to expiration is given by

$$\frac{\partial V}{\partial T}. \quad (\text{A.54})$$

In practice, it is often more useful to define the *theta* of an option as the change in the value of the option given the passage in time (and thus in a decrease in the time remaining in the life of the option). That is, the theta of an option is given by

$$\Theta = -\frac{\partial V}{\partial T}. \quad (\text{A.55})$$

The theta of an option can be approximated as follows:

$$\Theta \approx -\frac{V(T + \epsilon) - V(T)}{\epsilon}, \quad (\text{A.56})$$

where $V(T)$ represents the value of the option when the time until expiration is T . According to Black-Scholes, the theta for European calls and puts is found using Equation A.55:

$$\begin{aligned} \Theta_c &= -\frac{\partial c}{\partial T} = -\frac{Sn(d_1)\sigma}{2\sqrt{T}} - rXe^{-rT}N(d_2) \\ \Theta_p &= -\frac{\partial p}{\partial T} = -\frac{Sn(d_1)\sigma}{2\sqrt{T}} + rXe^{-rT}N(-d_2). \end{aligned}$$

Figure A.15 shows the theta of a call option according to the Black-Scholes formula. Theta is usually negative for options. Think about it: theta represents time decay. As time goes on, your option is running out. That is usually a bad thing. Can you think of an example of an option that has positive theta?³⁷

³⁷A very deep in-the-money put can have a positive theta. As an extreme example, consider a put on an underlying asset that has gone to zero. This is as good as it gets for the holder of a put option. In this case, the holder of a put option wants to exercise the option as soon as possible.

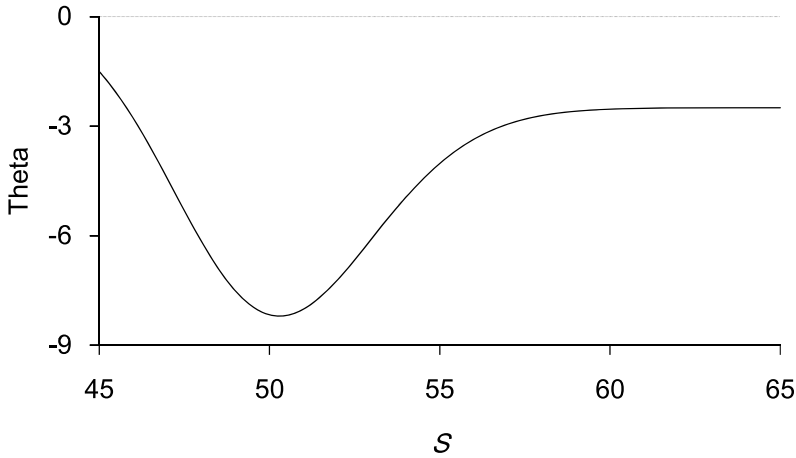


Figure A.15: Theta of Call Option as a Function of Underlying Asset When $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

A.7 Binary Options

Binary options, a type of exotic option, make their appearance in the swaps market relatively frequently. As such, we will spend a considerable amount of time discussing them. Binary options, also known as *digital options*, provide the holder with either nothing or a prespecified payment at expiration. This prespecified payment may consist of either a cash sum or a number of units of the underlying asset. The former type is known as a *cash-or-nothing option*. To take a concrete example, consider a cash-or-nothing call option in which the holder receives a payment of K if the underlying asset price is at least X at expiration and nothing otherwise. That is, the payoff from the option is:

$$\text{Payoff} = \begin{cases} K & \text{if } S_T \geq X \\ 0 & \text{if } S_T < X. \end{cases}$$

This payoff pattern is shown in Figure A.16.

Under the Black-Scholes assumptions value of the binary cash-or-nothing call option, c_{CN} , is³⁸

³⁸Theoretically, in the Black-Scholes model, there is no difference in price between a binary call option that pays K if the underlying asset price finishes at exactly the strike X at expiry and one that pays zero in this case (since the Black-Scholes model attributes zero probability to any one asset price obtaining). Practically speaking, option traders

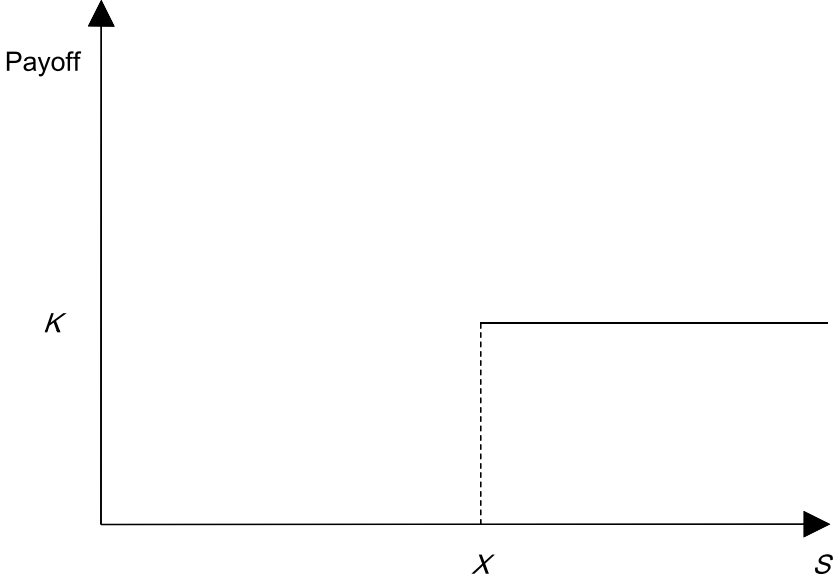


Figure A.16: Payoff of Cash-or-Nothing Binary Call Option

$$c_{CN} = Ke^{-rT}N(d_2). \quad (\text{A.57})$$

Similarly, with a cash-or-nothing put option, the holder receives a payment of K if the underlying asset price is at or below X at expiration and nothing otherwise. That is, the payoff from the option is:

$$\text{Payoff} = \begin{cases} K & \text{if } S_T \leq X \\ 0 & \text{if } S_T > X. \end{cases} \quad (\text{A.58})$$

This payoff pattern is shown in Figure A.17. Under the Black-Scholes assumptions, the value of the binary cash-or-nothing put option, p_{CN} , is

$$p_{CN} = Ke^{-rT}[1 - N(d_2)]. \quad (\text{A.59})$$

Figure A.18 shows the value of a binary cash-or-nothing put option as a function of the underlying.

are very concerned as to whether or not they are paying K if the asset price finishes on the strike, often charging different premia for the two, particularly if the strike price X is believed to be a “focal” point upon which it is thought the asset price may land.

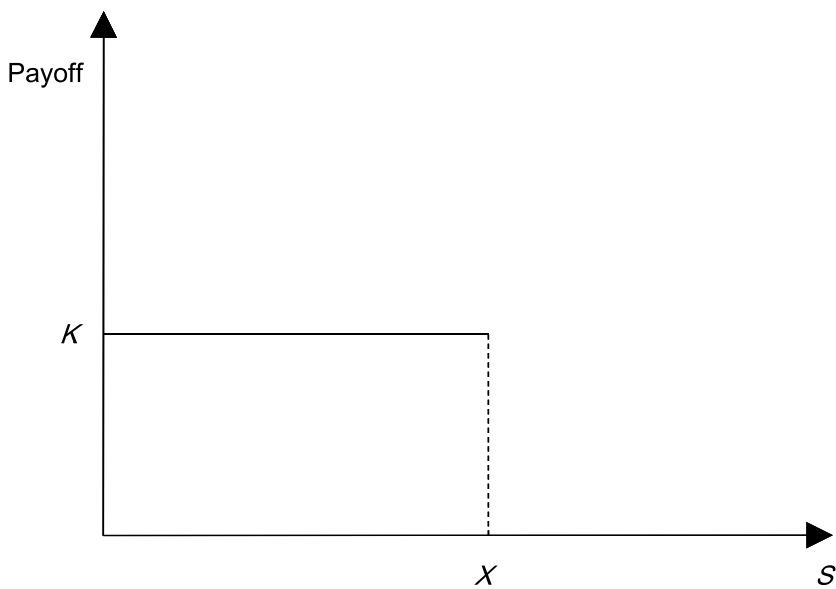


Figure A.17: Payoff of Cash-or-Nothing Binary Put Option

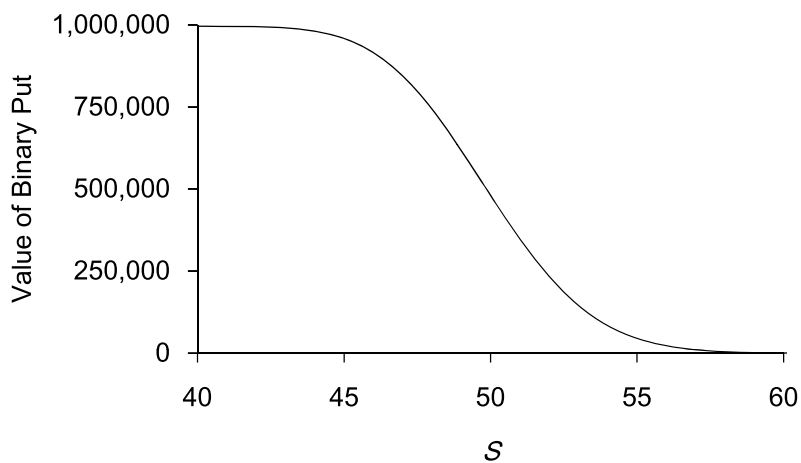


Figure A.18: Value of Cash-or-Nothing Binary Put Option as a Function of Underlying Asset When $K = 1\text{mm}$, $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

A.7.1 Delta of Binary Options

Binary options have some pretty strange properties, which we will examine. Let's start out by looking at the delta of binary options. As we discussed in Section A.6.1, the delta of an option, Δ , is the change in the option price with respect to a change in the underlying. Delta is typically the first sensitivity parameter that an option trader will look at, as changes in the underlying can happen quickly and significantly impact the value of the option. The delta of binary cash-or-nothing calls, denoted $\Delta(c_{CN})$, and binary cash-or-nothing puts, $\Delta(p_{CN})$, can be determined analytically from Equations A.57 and A.59, respectively:

$$\Delta(c_{CN}) = \frac{\partial c_{CN}}{\partial S} = \frac{Ke^{-rT}n(d_2)}{\sigma S\sqrt{T}}$$

$$\Delta(p_{CN}) = \frac{\partial p_{CN}}{\partial S} = -\frac{Ke^{-rT}n(d_2)}{\sigma S\sqrt{T}}.$$

The delta of a cash-or-nothing binary put is shown in Figure A.19. This figure looks a little weird compared to the delta of an ordinary put option as shown in Figure A.11. For an ordinary put option, the delta is always negative and monotonic; it is close to zero when the put is out-of-the-money and becomes more and more negative as the put goes further in-the-money. This means that if an option trader is long an ordinary put option and is delta

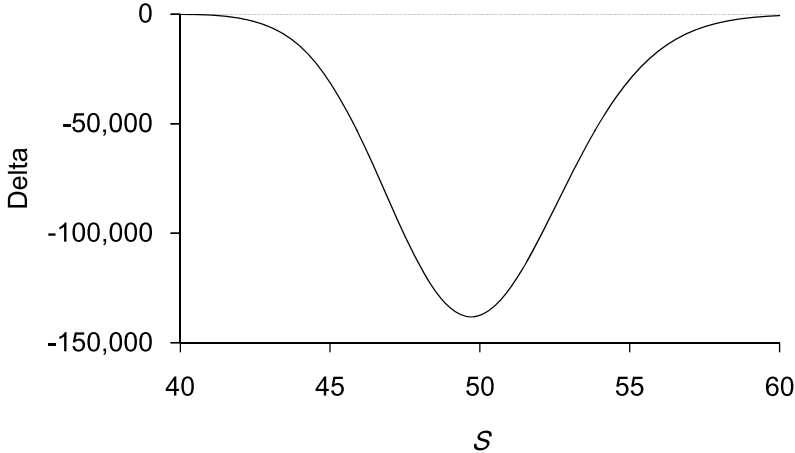


Figure A.19: Delta of a Cash-or-Nothing Binary Put Option as a Function of Underlying Asset When $K = 1\text{mm}$, $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

hedging the option, he will be long very little of the underlying when the option is deep out-of-the-money but will buy more and more of the underlying as the underlying price goes down and the option goes deeper in-the-money.

The situation is quite different with a binary put option. Consider the position of an option trader who is long a binary put and is delta hedging. If the put is deep out-of-the-money, the option trader will be long very little of the underlying. As the underlying price goes down, the trader will initially buy more and more of the underlying, similarly as he would if he were delta hedging an ordinary put option. However, as the binary put goes in-the-money, the delta profile of the option changes course dramatically, and the option trader actually starts selling some of his position in the underlying. As the underlying price continues to go down, and the binary put goes deeper and deeper in-the-money, the option trader will continue to sell the underlying until he again holds very little of the underlying as a hedge.

What's the rationale behind this seemingly schizophrenic delta hedging activity? When the underlying price is high and the binary put is deep out-of-the-money, the option trader will need to hold very little of the underlying, since any move downwards in the underlying doesn't increase the value of the option that much. As the underlying price goes down and approaches being at-the-money, the value of the binary put starts to go up significantly, and in order to remain delta neutral, the trader will buy more and more of the underlying (as the underlying goes down, the value of the shares of the underlying that the trader has purchased to delta hedge goes down, which offsets the increase in the value of the option). However, as the underlying continues to go down and the binary put option is in-the-money, there is no further upside in terms of the potential payout from the option. Although a further decline in the underlying price will further buffer against the possibility of the option finishing out-of-the-money, the payout from the option is fixed. As the underlying continues to fall, the value of the option isn't rising that much more, but the shares of the underlying that the trader is long to delta hedge continue to fall in value. This is markedly different from an ordinary put, in which any decreases in the underlying through the strike continue to increase the value of the put significantly. In order to remain delta neutral, the trader who is long this binary put option must actually start to sell some of the shares that he had been buying as the underlying declines below the strike.

The characteristics of the delta hedging strategy for the binary put option that was just described can be summarized by looking at its gamma profile as displayed in Figure A.20. In this figure, we have not only displayed the gamma of the option but we have also depicted the delta hedging activities

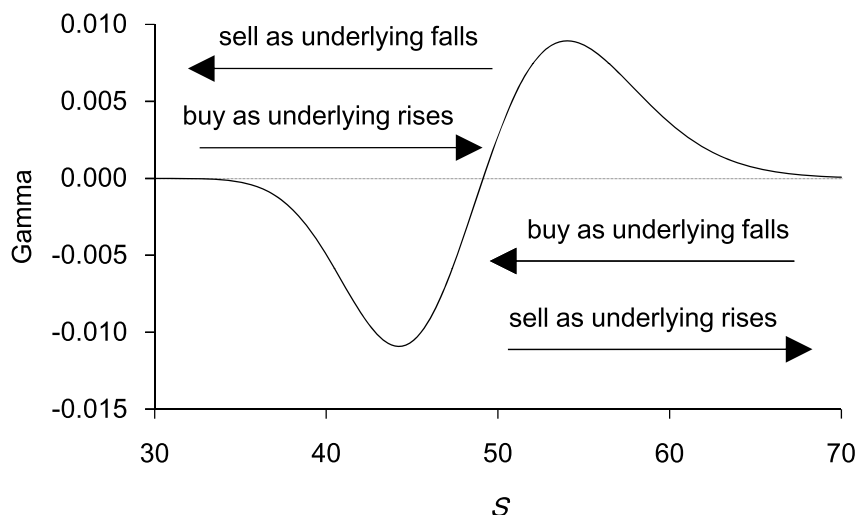


Figure A.20: Gamma of Cash-or-Nothing Binary Put as a Function of Underlying Asset, with Description of Delta Hedging Activities for a Long Position in the Option When $K = 1$, $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 3$ Month

of a trader who is long this option. When the underlying is above the strike and the option is out-of-the-money, the holder of the binary put is *long gamma* (meaning gamma is positive), and when the underlying is low and the option is in-the-money, the holder of the binary put is *short gamma* (meaning gamma is negative). When the option is out-of-the-money and its gamma is positive, a trader who is long the option and delta hedging winds up buying the underlying as it falls and selling the underlying as it rises. And when the option is in-the-money and its gamma is negative, a trader who is long the option and delta hedging winds up selling the underlying when it falls and buying the underlying as it rises. It is the change in the sign of the gamma of this option that is responsible for the erratic delta hedging pattern associated with it.

Potential Stabilizing or Destabilizing of the Underlying Through Delta Hedging

One important implication of the delta hedging activities of binary options is the potential stabilizing or destabilizing forces that such hedging can have on the price movements of the underlying. It is important to note that these phenomena would only potentially manifest themselves when these delta

hedging activities were large relative to other trading activities in the underlying itself. Whether or not delta hedging activities are stabilizing or destabilizing depends on whether the trader delta hedging is long or short the option and whether or not the option is in-the-money or out-of-the-money.³⁹ Consider, for example, a trader who is long a binary put option that is out-of-the-money and is delta hedging it. As the underlying goes down, the trader will buy the underlying, which on the margin will tend to drive the price of the underlying up. And as the underlying goes up, the trader will sell the underlying, which on the margin will tend to drive the price of the underlying down. Thus, the delta hedging activities of a trader who is long an out-of-the-money binary put tend to be a stabilizing force on the value of the underlying, just as the delta hedging activities by a trader who is long ordinary options can result in stabilizing the price of the underlying. If the delta hedging activities of such binary options are significant enough, this can be an overwhelming force in the market, resulting in the underlying being rather stable and trading in a tight range.

By comparison, if a trader is long a binary put that is in-the-money, the delta hedging activities of these options can result in tremendous destabilization of the underlying. If the binary put is in-the-money, then as the underlying price goes down, the trader who is long the option and delta hedging will sell some of the underlying, which will further serve to push the underlying down. And while the binary put is in-the-money, if the underlying price goes up, the trader will start buying some of the underlying, which will further serve to push the underlying up. Thus, the delta hedging activities for in-the-money binary puts can actually serve, at the margin, to destabilize the price of the underlying. Again, if the delta hedging activities of these options are large compared to outright trading in the underlying from other traders in the market, this can create destabilizing forces in the underlying, resulting in erratic price moves.⁴⁰

Delta Hedging Binary Options as Expiry Approaches

The delta of a binary option is relatively well behaved for a significant period of time before maturity. However, when the time to expiry is short, the delta

³⁹We assume here that only one of the two counterparties involved in the option trade will delta hedge. The other is assumed to hold the option to maturity without delta hedging. This is often the case when a dealer buys or sells an option to a buy-side account: the dealer often delta hedges the position and the client does not. If both counterparties delta hedge, then the delta hedging activities of each counterparty could offset one another.

⁴⁰This is a concept explored in Chapter 9, where we look at binary curve options. The delta hedging activities of in-the-money binary curve floors at the height of the financial crisis are widely regarded as having contributed to USD 30-year swap spreads going negative for the first time in history.

of a binary option is unstable if the option is near-the-money. Remember, the delta of the option measures the change of the value of the option given a change in the underlying. When the binary option is either deep in-the-money or deep out-of-the-money, the delta is small. Consider now a binary call option. If the binary is deep out-of-the-money, a slight increase in the underlying is going to help bring the option closer to the money. As with an ordinary option, this will increase the value of the option only slightly, and thus the delta is small. If the option is deep in-the-money, a slight increase in the underlying doesn't do much either. It looked like the owner was going to get K (the fixed binary payment of the option) before the move, and it still looks like he'll get K , with a slightly greater probability. Thus, for a deep in-the-money binary call option a slight change in the underlying does not change the value of the option much, and the delta is again small. It is when the option is near-the-money that the delta is greatest, as a slight fluctuation in the underlying most significantly changes the probability of receiving K instead of 0.

Figure A.21 shows the delta for a binary call option as a function of the underlying for several different expiries. As this figure shows, when time to expiration is short, like a day, the delta is out of control if we are near-the-money. In this case, a slight fluctuation in the underlying can mean the difference of getting 0 or getting K , which has been set to \$1mm. This is

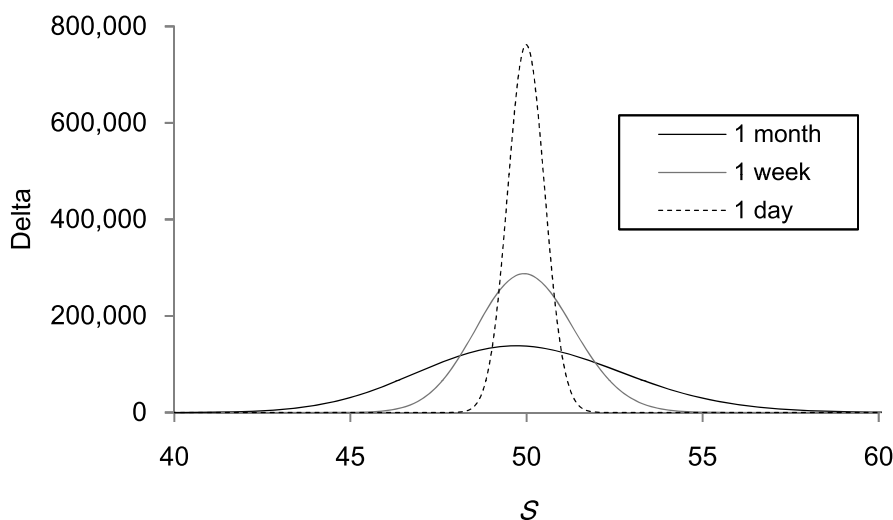


Figure A.21: Delta of a Cash-or-Nothing Binary Call as a Function of Underlying Asset for Various Expiries When $K = 1\text{mm}$, $X = 50$, $r = 5\%$, $\sigma = 20\%$